

# Zhenyuan Zhang

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## EDUCATION

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### PhD candidate in Human Movement Biomechanics

Jul 2023 – Present

Research Institute for Sport and Exercise Science  
Liverpool John Moores University, United Kingdom

### Master of Science in Sport and Clinical Biomechanics

Sep 2020 – Nov 2021

School of Sport and Exercise Science  
Liverpool John Moores University, United Kingdom

### Bachelor of Education in Human Movement Science

Sep 2016 – Jun 2020

School of Exercise and Health Science  
Chengdu Sport University, China

## SKILLS

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**Languages:** Python, Matlab, Git, Shell, L<sup>A</sup>T<sub>E</sub>X

**Computer Simulations:** OpenSim, Visual3D, Pyomeca, MyoSuite

**Biomechanics Tools:** Optical Motion Capture, Inertial Measurement Units, Force Plates, Electromyography

**High-Performance Computing:** Slurm, AWS, Dask

**Machine Learning:** PyTorch, TensorFlow, Keras, Scikit-learn

## TECHNICALITIES

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### Machine Learning and Biomechanics Researcher

Jul 2023 – Present

SportScientia Ltd., Industry Partnership  
Remote

- Developed and validated deep learning neural network models to estimate ground reaction forces from smart insoles instrumented with pressure sensors and inertial measurement units for various movement tasks.
- Validated smart insoles with customized sensor fusion algorithm for measuring spatiotemporal gait parameters against optical motion capture and force plates for athletic performance and load monitoring applications.
- Assisted in developing cloud computing pipelines with AWS to automate data processing and analysis for the smart insole system.

### Biomechanics Laboratory Technician

Sep 2025 – Jan 2026

Liverpool John Moores University, Contracted  
United Kingdom

- Built up and fine-tuned 3 advanced biomechanics systems with the senior technician, provided professional training for staff and students. I am also responsible for managing them for both teaching and research activities.
- System1: 10 Qualysis<sup>®</sup> Arqus motion capture cameras integrated with 8 Qualysis<sup>®</sup> Miquis cameras for marker-less motion capture, 2 Kistler<sup>®</sup> force plates, 16 Delsys<sup>®</sup> Trigno EMGs.
- System2: 8 Qualysis<sup>®</sup> Arqus motion capture cameras integrated with a Treadmetrix<sup>®</sup> treadmill (AMTI<sup>®</sup> force plate embedded), 16 Delsys<sup>®</sup> Trigno EMGs and 8 Noraxon<sup>®</sup> EMGs.
- System3: 14 Vicon<sup>®</sup> Vero motion capture cameras integrated with 16 Vicon<sup>®</sup> T-series cameras, 2 Kistler<sup>®</sup> force plates, 8 Vicon<sup>®</sup> Blue Trident IMUs and 16 Delsys<sup>®</sup> Trigno EMGs.

### Graduate Research Assistant

Nov 2021 – Jul 2023

Liverpool John Moores University, Contracted  
United Kingdom

- Assisted in commercial projects with New Balance Athletics, USA to test biomechanical interactions between soccer boots with different studs and artificial turfs using high-speed motion capture and force plates.
- Assisted in commercial projects with New Balance Athletics, USA to test effects of different running shoes on lower limb biomechanics and muscle co-contractions during treadmill running using motion capture integrated with instrumented treadmill and EMGs.

## PUBLICATIONS

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- Zhang, Z.**, Verhuel, J., Robinson, M., and Lake, M. Estimating ground reaction forces in dynamic sports movements using instrumented insoles and deep learning. *Oral Presentation at XXX Congress of International Society of Biomechanics*, p.85033, Stockholm, Sweden. (2025)
- Yang, C., Yang, Y., Xu, Y., **Zhang, Z.**, Lake, M., and Fu, W. Whole leg compression garments influence lower limb kinematics and associated muscle synergies during running. *Frontiers in Bioengineering and Biotechnology*, 12, 1310464. (2024)
- Zhang, Z.** and Lake, M. Rate of knee flexion at the instant of landing during running can influence initial knee joint stiffness estimates due to running shoe cushioning. *Oral Presentation at XXIX Congress of International Society of Biomechanics*, p.314, Fukuoka, Japan. (2023)
- Zhang, Z.** and Lake, M. A re-examination of the measurement of foot strike mechanics during running: the immediate effect of footwear midsole thickness. *Frontiers in Sports and Active Living*, 4, 824183. (2022)
- Zhang, Z.** and Lake, M. A comparison of unmatched and matched filtering approaches for knee joint stiffness calculation during running. *Oral Presentation at 40<sup>th</sup> International Society of Biomechanics in Sports, Proceedings Archive*, 40(1), 807, Liverpool, United Kingdom. (2022)

## PROJECTS

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### TechLayer | [GitHub \(Coming Soon\)](#)

- A Python project which trained and implemented a deep learning model to predict Ground Reaction Forces (GRFs) from IMU and pressure sensor data collected from instrumented insoles during various dynamic sports movements.
- The model is trained and validated on a dataset of 32 participants using High-Performance Computing (HPC) clusters and demonstrated high accuracy in predicting vertical and anterior-posterior GRFs against force plates.

### Wearable\_IK | [GitHub \(Coming Soon\)](#)

- A Python project integrating open-source sensor fusion algorithms and OpenSim API to estimate joint kinematics from IMU data and validating the results against optical motion capture data.
- It automates the entire workflow from data loading, preprocessing, sensor fusion, sensor-to-segment calibration, inverse kinematics, to results visualization
- It also features parallelized computing to speed up processing and a calibration-free approach for IMU sensors as it does not require magnetometers.

### Wearable\_System | [GitHub \(Under Development\)](#)

- A Python project integrating **TechLayer** and **Wearable\_IK** to generate muscle-driven simulations of human movement while solving optimal control problems from wearable sensor (instrumented insoles + IMUs) using OpenSim Moco. It also aims to validate the simulation results against traditional biomechanics measurement system using optical motion capture, force plates and EMGs.

### My\_Website | [GitHub](#)

- A project to build and deploy my personal academic website from an open-source TypeScript template for fun.

## TEACHING

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### Graduate Teaching Assistant

Nov 2021 – Nov 2025

Liverpool John Moores University, Contracted  
United Kingdom

- Assisted in biomechanics lectures and practical laboratory sessions for optical motion capture cameras, force plates, IMUs and EMGs at undergraduate and master levels.
- Provided one-on-one tutorial support for students' projects and technical training of setting up and using the equipment for data collection of students' projects.

## SCHOLARSHIPS

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**University-Industry Matched PhD Fund:** A three-year matched funding package awarded jointly by Liverpool John Moores University and industry partner to support collaborative PhD research. (Jul 2023)

**LJMU International Achievement Scholarship:** A scholarship awarded to international students with excellent academic performance to pursue postgraduate studies at Liverpool John Moores University. (Sep 2020)

## REFERENCES

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References available upon request.